Practice Problem 8.4

The circuit in Fig. 8.12 has reached steady state at $t = 0^-$. If the make-before-break switch moves to position b at $t = 0$, calculate $i(t)$ for $t > 0$.

Answer: $e^{-2.5t}(10\cos 1.6583t - 15.076\sin 1.6583t)$ A.

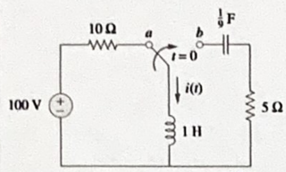

Figure 8.12
For Practice Prob. 8.4.

串联RLC二阶电路的零输入响应

① 求初始值 $i(0)=10A$ $\quad L\frac{di(0)}{dt}=-50 \Rightarrow \frac{di(0)}{dt}=-50$

② $\alpha=\frac{R}{2L}=2.5$, $\omega_0=\frac{1}{\sqrt{LC}}=3$ $\quad \alpha<\omega_0$ 欠阻尼

$\therefore \omega_d=\sqrt{\omega_0^2-\alpha^2}=1.6583$

通解: $e^{-\alpha t}(A_1\cos\omega_d t + A_2\sin\omega_d t)$

$i(0)=10 \Rightarrow A_1=10$

$\frac{di(0)}{dt}=-50 \Rightarrow -\alpha A_1 + A_2\omega_d=-50 \Rightarrow A_2=-15.076$

56 $\quad \therefore i(t)=e^{-2.5t}(10\cos 1.6583t - 15.076\sin 1.6583t)$ A

Practice Problem 8.6

Refer to the circuit in Fig. 8.17. Find $v(t)$ for $t > 0$.

Answer: $50(e^{-10t} - e^{-2.5t})$ V.

并联RLC二阶电路的零输入响应

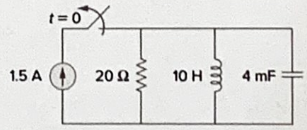

Figure 8.17

① 初始值: $V(0)=0$

$C\frac{dv(0)}{dt}=-1.5$ (∵20Ω两端电压为0,电流也为0) $\Rightarrow \frac{dv(0)}{dt}=-375$

② $\alpha=\frac{1}{2RC}=6.25$ $\quad \omega_0=\frac{1}{\sqrt{LC}}=5$ $\quad \alpha>\omega_0$ 过阻尼. $s_1=-2.5$ $\quad s_2=-10$

通解 $A_1e^{-2.5t}+A_2e^{-10t}$

代入初始值,得到 $A_1=-50$, $A_2=50$

57 $\quad \therefore V(t)=50e^{-10t}-50e^{-2.5t}$ V

Practice Problem 8.7

Having been in position a for a long time, the switch in Fig. 8.21 is moved to position b at $t = 0$. Find $v(t)$ and $v_R(t)$ for $t > 0$.

串联RLC的阶跃响应（全响应）

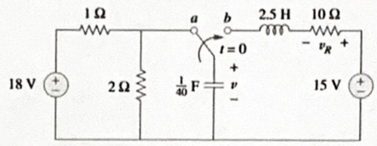

Figure 8.21
For Practice Prob. 8.7.

Answer: $15 - (1.7321 \sin 3.464t + 3 \cos 3.464t)e^{-2t}$ V, $3.464e^{-2t} \sin 3.464t$ V.

$V(0) = 12$

$$\begin{cases} C\dfrac{dv(0)}{dt} = 0 \Rightarrow \dfrac{dv(0)}{dt} = 0 \\ V(\infty) = 15 \end{cases}$$

$\alpha = \frac{R}{2L} = 2 \quad \omega_0 = \frac{1}{\sqrt{LC}} = 4$

$\therefore$ 欠阻尼 $\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 3.464$

通解：$15 + e^{-2t}(A_1 \cos\omega_d t + A_2 \sin\omega_d t)$

代入初始值 $\Rightarrow A_1 = -3;\ A_2 = -1.732$

$\therefore v(t) = 15 - e^{-2t}(3\cos 3.464t + 1.732\sin 3.464t)$ V

$i = C\frac{dv}{dt}, \quad V_R = 10i = \frac{1}{4}\frac{dv}{dt}$

$= 3.464\, e^{-2t} \cdot \sin 3.464t$ V

58

Determine v and i for $t > 0$ in the circuit of Fig. 8.28. (See comments about current sources in Practice Prob. 7.5.)

Practice Problem 8.9

Answer: $20(1 - e^{-5t})$ V, $5(1 - e^{-5t})$ A.

二阶电路的通用解法

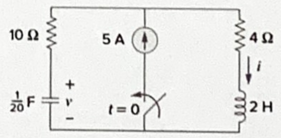

Figure 8.28
For Practice Prob. 8.9.

① 初始值及稳态值
$$\begin{cases} V(0) = 0 \\ C\dfrac{dv(0)}{dt} = 5 \Rightarrow \dfrac{dv(0)}{dt} = 100 \\ V(\infty) = 20 \end{cases}$$

② Turn off 独立源
$$\begin{cases} -i = \dfrac{1}{20}\dfrac{dv}{dt} \\ -v + 4i + 2\dfrac{di}{dt} = 0 \end{cases} \Rightarrow \frac{d^2v}{dt^2} + 7\frac{dv}{dt} + 10 = 0 \Rightarrow \begin{matrix} s_1 = -2 \\ s_2 = -5 \end{matrix}$$

③ 通解：$V(t) = 20 + A_1 e^{-2t} + A_2 e^{-5t}$

59

代入初始值 $\Rightarrow A_1 = 0,\ A_2 = -20$

$\therefore V(t) = 20 - 20e^{-5t}$ V

$i(t) = 5 - C\frac{dv}{dt}$

$= 5 - 5e^{-5t}$ A